

Prospectus gap-closure results

The campaign covers 32 bonds: 16 qualified positives, 16 qualified negatives and 0 unresolved answers. A positive identifies a principal write-down or compulsory conversion into common shares in the selected terms. A negative is limited to the examined documents and supported constructions.

The full prospectus and bank regression suite passed 734 tests. All 32 integrated cases retained all 14 bank obligations per case. Actual customer eligibility and transaction permission remain unestablished.

Disposition of the seven gaps

Gap	Result	Remaining requirement
1. Monetary fields / Enel	Fixed in the bounded reader. Exact source-field and rational checks support Enel's EUR 1,000 unit/redemption relation.	New layouts and unit conventions require additional construction tests.
2. Unfamiliar legal wording	Added loss-rights/exception guards, independent money checks, tampering tests, and repaired the fresh quoted-heading defect.	Unrestricted English entailment and all possible shared interpretation errors remain unproved.
3. Dependencies and editions	Quoted references and dated source selections are checked; three named Enel/Unilever source links inspected.	Unilever/Enel separate agreements, incorporated materials, amendments and the Unilever trust-deed date discrepancy remain open.
4. Authority and language	Deutsche German/English section 5(4) examined locally with exact sources; Lloyds official routes retried.	Lloyds official original was not obtained; whole-document translation equivalence and current authority remain unverified.
5. Mechanism detail	Each positive witness retains the full conditional quotation. Trigger, amount, restoration, notice and settlement requirements remain explicit. The UBS-specific engine is connected.	Complete loss calculations for other issues, plus actual event, notice, holding and price facts.
6. Bank rules	Actual bank investigations retain all 14 obligations. Tests preserve missing, stale and conflicting facts and keep the feature separate from regulatory scope.	Dated customer, authority, bank policy, mandate, suitability and current-law evidence for a real investigation.
7. Unfamiliar families	Frozen challenge obtained KBC and Intesa. Source review found a witness defect, which was repaired and replayed.	Corporate challenge 0/2 after 40 retrieval attempts. Repaired candidate has no unseen-family observations; broader validation remains open.

Evidence and decision

Decision requirement	Finding
Decision	Retain the bounded reader repairs and conditional bank integration
Primary criterion	Constructed obligations, source replay, regression and formal/native checks pass; all 14 bank obligations are retained
Veto status	Invalid witnesses found in the first frozen challenge were removed; failed attempts remain archived
Main uncertainty	English interpretation, omitted contracts, source authority and actual customer facts
Next justified action	Obtain missing sources and conduct a further unseen-family challenge with the repaired method
Not concluded	Universal legal accuracy, complete contract reconstruction or permission to trade

Inference question	Status
Hard veto screen	Rejected forged amounts, conditions, source identities, scope shortcuts and stale facts; failed development attempts retained
Statistically supported ranking	None; deterministic finite checks and a small source challenge
Descriptive differences	Supported row counts and acquisition coverage only
Default readiness	Bounded engineering repair; no claim of universal legal classification
Next evidence needed	Referenced agreements and amendments, controlling-source review, more independently selected source families, actual bank/customer facts

Source findings

Enel's final terms name the base prospectus of 20 December 2024 and supplement of 27 January 2025; both retained editions were checked. The final redemption amount is EUR 1,000 per EUR 1,000 Calculation Amount. The base prospectus, p.95, makes the terms subject to separate agency provisions and identifies the agency, guarantee and covenant agreements available for inspection through the agents. Unilever's final terms name the retained memorandum of 16 May 2025. Its summary refers to a trust deed dated 22 July 1994 (p.8), while the operative conditions identify a trust deed dated 16 May 2024 (p.32). Amendment history may explain this, but the executed agreements have not been obtained. The paying agency agreement is separately identified as 22 July 1994, as amended. Deutsche Bank's German terms control (p.4). The German and English columns of section 5(4), pp.52-55, both specify reduction when consolidated CET1 falls below 5.125%, pro-rata allocation and caps, prompt determination within one month unless shortened, and notice rules under which missing notice does not invalidate the reduction. Restoration is discretionary and subject to profit, distribution and continuing-trigger constraints. This local reading does not establish translation equivalence for the whole prospectus. Lloyds' SEC route returned HTTP403. The issuer URL returned HTTP200 with an HTML error page, not a prospectus PDF. The existing mirror remains explicitly qualified. The first frozen challenge returned positive answers for KBC and Intesa,

supported by actual write-down clauses. Inspection also found invalid supporting witnesses in a quoted heading and an uncertainty statement. Three constructed counterexamples failed before repair and passed afterward. The original frozen result is preserved; both cases are now exposed regression evidence. The acquisition ledger records all 40 retrieval attempts, including failures, discovery pages, two rejected PDF candidates and source-selection decisions. No corporate issue bundle was obtained. Source availability is therefore a material limit on the challenge, and there is no population-accuracy claim.

Bond results

1. Tesco treasury 3.500% senior notes due 2033

Identity: XS3201918409. **Answer:** No.

No feature identified within the examined offering terms: cash principal repayment is provided (tesco-2025-final, PDF pp.2-4); no applicable principal write-down or compulsory common-share conversion was identified. This is a qualified source reading, not a proof of absence in all operative language or documents.

Selected sources: [tesco-2025-final](#); [tesco-2025-base](#); [tesco-2025-supplement](#).

Offering-circular conditions and selected final terms, as supplemented. Financial statements and the separate trust/agency agreements are not reconstructed as complete legal contracts.

2. Veolia 3.690% senior notes due 2031

Identity: FR0014017P12. **Answer:** No.

No feature identified within the examined offering terms: cash principal repayment is provided (veolia-2026-base, PDF p.79); no applicable principal write-down or compulsory common-share conversion was identified. This is a qualified source reading, not a proof of absence in all operative language or documents.

Selected sources: [veolia-2031-final](#); [veolia-2026-base](#).

Dated final terms and base-prospectus disclosures. Separate agency agreement, corporate filings and subsequent changes are not proved complete.

3. Veolia 4.122% senior notes due 2036

Identity: FR0014017P04. **Answer:** No.

No feature identified within the examined offering terms: cash principal repayment is provided (veolia-2026-base, PDF p.79); no applicable principal write-down or compulsory common-share conversion was identified. This is a qualified source reading, not a proof of absence in all operative language or documents.

Selected sources: [veolia-2036-final](#); [veolia-2026-base](#).

Dated final terms and base-prospectus disclosures. Separate agency agreement, corporate filings and subsequent changes are not proved complete.

4. Deutsche Bank 6.75% perpetual AT1, December 2025

Identity: DE000A460DG7. **Answer:** Yes.

Yes: the principal can be reduced or cancelled under the contractual terms (deutsche-at1-2025, PDF p.9). The conditions are retained in the cited evidence; the power need not have been exercised.

Selected sources: [deutsche-at1-2025](#).

German terms prevail; English translation and two-column extraction are not independently certified equivalent.

5. Sogécap 6.25% perpetual restricted tier 1, July 2025

Identity: FR00140112W8. **Answer:** Yes.

Yes: the principal can be reduced or cancelled under the contractual terms (sogecap-rt1-2025, PDF p.72). The conditions are retained in the cited evidence; the power need not have been exercised.

Selected sources: [sogecap-rt1-2025](#).

6. UBS 5.600% SGD perpetual AT1 (June issue and July tap)

Identity: CH1357852636. **Answer:** Yes.

Yes: the bond can be compulsorily converted into common/ordinary shares under the contractual terms (ubs-sgd-at1-2024-final-published, PDF p.28). The conditions are retained in the cited evidence; the power need not have been exercised.

Selected sources: [ubs-sgd-at1-2024-final-published](#).

7. Barclays 7.625% perpetual AT1, first reset 2035

Identity: US06738EDC66. **Answer:** Yes.

Yes: the bond can be compulsorily converted into common/ordinary shares under the contractual terms (barclays-at1-2025, PDF p.57). The conditions are retained in the cited evidence; the power need not have been exercised.

Selected sources: [barclays-at1-2025](#).

8. HSBC 6.750% perpetual AT1, first reset 2031

Identity: hsbc-at1-2026-2031. **Answer:** Yes.

Yes: the bond can be compulsorily converted into common/ordinary shares under the contractual terms (hsbc-at1-2026-specific, PDF p.1). The conditions are retained in the cited evidence; the power need not have been exercised.

Selected sources: [hsbc-at1-2026-specific](#).

9. HSBC 7.000% perpetual AT1, first reset 2036

Identity: hsbc-at1-2026-2036. **Answer:** Yes.

Yes: the bond can be compulsorily converted into common/ordinary shares under the contractual terms (hsbc-at1-2026-specific, PDF p.1). The conditions are retained in the cited evidence; the power need not have been exercised.

Selected sources: [hsbc-at1-2026-specific](#).

10. Standard Chartered 7.625% perpetual AT1, January 2025

Identity: US853254DF47, USG84228GP72. **Answer:** Yes.

Yes: the principal can be reduced or cancelled under the contractual terms (standard-chartered-at1-2025-executed-deed, PDF pp.42-43). The conditions are retained in the cited evidence; the power need not have been exercised.

Selected sources: [standard-chartered-at1-2025-executed-deed](#).

Issuer-linked execution-version trust deed supplies final conditions. The retained offering circular has a subject-to-completion warning and remains excluded from final-term evidence.

11. Apple Inc. 4.000% Notes due 2028

Identity: apple-senior-2025.1. **Answer:** No.

No feature identified within the examined offering terms: cash principal repayment is provided (apple-senior-2025, PDF p.16); no applicable principal write-down or compulsory common-share conversion was identified. This is a qualified source reading, not a proof of absence in all operative language or documents.

Selected sources: [apple-senior-2025](#).

12. Apple Inc. 4.200% Notes due 2030

Identity: apple-senior-2025.2. **Answer:** No.

No feature identified within the examined offering terms: cash principal repayment is provided (apple-senior-2025, PDF p.16); no applicable principal write-down or compulsory common-share conversion was identified. This is a qualified source reading, not a proof of absence in all operative language or documents.

Selected sources: [apple-senior-2025](#).

13. Apple Inc. 4.500% Notes due 2032

Identity: apple-senior-2025.3. **Answer:** No.

No feature identified within the examined offering terms: cash principal repayment is provided (apple-senior-2025, PDF p.16); no applicable principal write-down or compulsory common-share conversion was identified. This is a qualified source reading, not a proof of absence in all operative language or documents.

Selected sources: [apple-senior-2025](#).

14. Apple Inc. 4.750% Notes due 2035

Identity: apple-senior-2025.4. **Answer:** No.

No feature identified within the examined offering terms: cash principal repayment is provided (apple-senior-2025, PDF p.16); no applicable principal write-down or compulsory common-share conversion was identified. This is a qualified source reading, not a proof of absence in all operative language or documents.

Selected sources: [apple-senior-2025](#).

15. Duke Energy Corporation 5.45% Senior Notes due 2034

Identity: duke-senior-2024.1. **Answer:** No.

No feature identified within the examined offering terms: cash principal repayment is provided (duke-senior-2024, PDF p.15); no applicable principal write-down or compulsory common-share conversion was identified. This is a qualified source reading, not a proof of absence in all operative language or documents.

Selected sources: [duke-senior-2024](#).

16. Duke Energy Corporation 5.80% Senior Notes due 2054

Identity: duke-senior-2024.2. **Answer:** No.

No feature identified within the examined offering terms: cash principal repayment is provided (duke-senior-2024, PDF p.15); no applicable principal write-down or compulsory common-share conversion was identified. This is a qualified source reading, not a proof of absence in all operative language or documents.

Selected sources: [duke-senior-2024](#).

17. The Southern Company Series 2025A 6.50% Junior Subordinated Notes due 2085

Identity: southern-junior-2025a.1. **Answer:** No.

No feature identified within the examined offering terms: cash principal repayment is provided (southern-junior-2025a, PDF p.10); no applicable principal write-down or compulsory common-share conversion was identified. This is a qualified source reading, not a proof of absence in all operative language or documents.

Selected sources: [southern-junior-2025a](#).

18. Duke Energy Corporation 6.45% Fixed-to-Fixed Reset Rate Junior Subordinated Debentures due 2054

Identity: duke-junior-2024.1. **Answer:** No.

No feature identified within the examined offering terms: cash principal repayment is provided (duke-junior-2024, PDF p.12); no applicable principal write-down or compulsory common-share conversion was identified. This is a qualified source reading, not a proof of absence in all operative language or documents.

Selected sources: [duke-junior-2024](#).

19. NatWest Markets 4.789% senior notes due 2028

Identity: USG6382RGD47, US63906YAM03. **Answer:** Yes.

Yes: the principal can be reduced or cancelled under a disclosed statutory resolution power (nwm-2025-base, PDF p.69). The conditions are retained in the cited evidence; the power need not have been exercised.

Selected sources: [nwm-2025-final](#); [nwm-2025-base](#); [nwm-2025-registration](#).

20. BPCE USD SOFR + 1.03% senior preferred notes due 2030

Identity: FR0014014YQ1. **Answer:** Yes.

Yes: the principal can be reduced or cancelled under a disclosed statutory resolution power (bpce-2025-base, PDF p.150). The conditions are retained in the cited evidence; the power need not have been exercised.

Selected sources: [bpce-2025-23-final](#); [bpce-2025-base](#).

21. CBA EUR ESTR + 0.26% senior notes due October 2026

Identity: XS3206613666. **Answer:** No.

No feature identified within the examined offering terms: cash principal repayment is provided (cba-2025-6700-final, PDF pp.2-4); no applicable principal write-down or compulsory common-share conversion was identified. This is a qualified source reading, not a proof of absence in all operative language or documents.

Selected sources: [cba-2025-6700-final](#); [cba-2025-base](#); [cba-2025-emtn-supplement](#).

Dated offering-circular disclosures and selected supplement. The separate agency/deed documents and later law are outside the source conclusion.

22. Compass Group 3.500% senior notes due 15 January 2035

Identity: XS3232968985. **Answer:** No.

No feature identified within the examined offering terms: cash principal repayment is provided (compass-2035-final, PDF pp.2-3); no applicable principal write-down or compulsory common-share conversion was identified. This is a qualified source reading, not a proof of absence in all operative language or documents.

Selected sources: [compass-2035-final](#); [compass-2025-base](#); [compass-2025-supplement](#).

Dated offering disclosures and selected supplement only; incorporated financial filings and complete separate trust/agency agreements are not reconstructed.

23. ING Groep 7.000% perpetual AT1, first reset November 2032

Identity: US456837BT90. **Answer:** Yes.

Yes: the bond can be compulsorily converted into common/ordinary shares under the contractual terms (ing-at1-us456837bt90, PDF p.36). The conditions are retained in the cited evidence; the power need not have been exercised.

Selected sources: [ing-at1-us456837bt90](#).

24. Shell International Finance 1.000% senior notes due 10 December 2030

Identity: XS2228291279. **Answer:** No.

No feature identified within the examined offering terms: cash principal repayment is provided (shell-2030-final, PDF pp.2-3); no applicable principal write-down or compulsory common-share conversion was identified. This is a qualified source reading, not a proof of absence in all operative language or documents.

Selected sources: [shell-2030-final](#); [shell-2020-base](#).

Dated offering disclosures only. Separate complete trust deeds, subsequent issuer substitutions and amendments are not reconstructed.

25. Danske Bank DKK1.25bn perpetual AT1, Series 4, August 2026

Identity: DK0030577200. **Answer:** Yes.

Yes: the bond can be compulsorily converted into common/ordinary shares under the contractual terms (danske-at1-2026-base, PDF p.110). The conditions are retained in the cited evidence; the power need not have been exercised.

Selected sources: [danske-dkk-at1-2026-final](#); [danske-at1-2026-base](#).

26. Lloyds 6.625% perpetual AT1, first reset September 2035

Identity: US539439BF59. **Answer:** Yes.

Yes: the bond can be compulsorily converted into common/ordinary shares under the contractual terms (lloyds-at1-2025-yahoo-filing, PDF p.1). The conditions are retained in the cited evidence; the power need not have been exercised.

Selected sources: [lloyds-at1-2025-yahoo-filing](#).

27. Banco Santander 6.000% perpetual AT1, July 2025

Identity: XS3100756637. **Answer:** Yes.

Yes: the bond can be compulsorily converted into common/ordinary shares under the contractual terms (santander-at1-2025, PDF p.103). The conditions are retained in the cited evidence; the power need not have been exercised.

Selected sources: [santander-at1-2025](#).

Feature reading of this dated circular. Incorporated financial reports, all applicable law and later amendments are not a proved closed set.

28. Unilever Capital Corporation 2.750% notes due 22 May 2030

Identity: XS3081333547. **Answer:** No.

No feature identified within the examined offering terms: cash principal repayment is provided (unilever-2030-final, PDF pp.2-4); no applicable principal write-down or compulsory common-share conversion was identified. This is a qualified source reading, not a proof of absence in all operative language or documents.

Selected sources: [unilever-2030-final](#); [unilever-2025-base](#).

Feature reading of the 20 May 2025 final terms and expressly named 16 May 2025 Information Memorandum. The separate 16 May 2024 Trust Deed, constitutional documents and financial reports incorporated by reference, later amendments and all applicable law are not independently reconstructed. A negative is bounded to the retained offering disclosures, not a proved absence across the complete legal contract.

29. ABN AMRO 5.750% perpetual AT1, February 2025

Identity: XS3004202811. **Answer:** Yes.

Yes: the principal can be reduced or cancelled under the contractual terms (abn-at1-2025, PDF p.1). The conditions are retained in the cited evidence; the power need not have been exercised.

Selected sources: [abn-at1-2025](#).

Feature reading of the dated standalone offering circular. Incorporated financial reports, separate agreements, later amendments and applicable law are not a proved closed set.

30. Enel Finance International 2.625% senior notes due 24 February 2028

Identity: XS3008888953. **Answer:** No.

No feature identified within the examined offering terms: cash principal repayment is provided (enel-2025-final, PDF pp.4-7); no applicable principal write-down or compulsory common-share conversion was identified. This is a qualified source reading, not a proof of absence in all operative language or documents.

Selected sources: [enel-2025-final](#); [enel-2024-base](#); [enel-2025-supplement](#).

Dated final terms, selected 20 December 2024 base and 27 January 2025 supplement only. Financial and constitutional documents, separate agency/guarantee/covenant agreements, later amendments and complete applicable law remain outside proved source closure.

31. KBC Group undated EUR 1 billion AT1, May 2025

Identity: BE0390219856. **Answer:** Yes.

Yes: the principal can be reduced or cancelled under the contractual terms (kbc-at1-2025, PDF p.57). The conditions are retained in the cited evidence; the power need not have been exercised.

Selected sources: [kbc-at1-2025](#).

Standalone issue terms examined; incorporated financial reports, separate agency agreement and later law/amendments not completely reconstructed.

32. Intesa Sanpaolo EUR 750 million 5.500% AT1, February 2026

Identity: XS3298382931. **Answer:** Yes.

Yes: the principal can be reduced or cancelled under the contractual terms (intesa-at1-2026, PDF p.86). The conditions are retained in the cited evidence; the power need not have been exercised.

Selected sources: [intesa-at1-2026](#).

Issue-specific NC6 conditions and common disclosures examined; referenced financial/EMTN information, agency agreement, current law and amendments not completely reconstructed.

Limits and next evidence

The bounded parser and integration repairs are implemented and checked. Complete contracts, current legal authority, unrestricted language interpretation and actual bank/customer facts remain open. The planned four-case source challenge is incomplete: two capital families were inspected, while no corporate bundle was obtained within the retrieval budget. A further unseen-source challenge is needed after the final witness repair.

The strongest alternative explanation for a passing software run is that extractor and checker share an incomplete interpretation. The source challenge is too small to establish accuracy in an issuer population. An applicable omitted condition, wrong edition, or accepted forged witness would overturn the affected result.